

Business Requirement Document

Project: UniVerse (Social Media Platform)

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1. Introduction

This document describes the business requirements for the development of a **Social Media Application**. The application aims to provide users with a platform to create profiles, connect with other users, and share content in the form of text and images.

The purpose of this Business Requirements Document (BRD) is to clearly define:

- Business goals
- User needs
- System scope
- Expected outcomes

This document serves as a reference for stakeholders, project managers, and the development team to ensure a shared understanding of the business expectations before system design and implementation begin.

2. Business Objectives

The primary business objectives of the Social Media Application are:

- To provide users with a digital platform to connect and interact with others
- To enable users to share content easily in a user-friendly environment
- To increase user engagement through social interactions such as likes, shares, and follows
- To deliver a personalized content experience based on user connections
- To ensure scalability and maintainability for future feature enhancements
- To enable real-time communication between users through direct messaging
- To keep users informed and engaged through in-app notifications
- To improve user retention by providing timely interaction updates

3. Scope of the Project

3.1 In Scope

The following functionalities are included within the scope of this project:

- User registration and secure login functionality

- User profile creation, update, and management
- Uploading and displaying profile pictures
- Following and unfollowing other users
- Creating posts with text and images
- Liking and sharing posts
- Viewing a personalized feed containing posts from followed users
- One-to-one direct messaging between users
- Viewing chat history between users
- In-app notifications for likes, shares, follows, and messages
- Notification status tracking (read/unread)

3.2 Out of Scope

- Video streaming
- Advertisement management
- Payment or subscription systems

4. Stakeholders

The key stakeholders involved in this project are:

- **Business Owner:** Defines business goals and expectations
- **Product Manager:** Oversees feature prioritization and requirement alignment
- **Development Team:** Responsible for system design and implementation
- **Quality Assurance (QA) Team:** Ensures system quality and requirement validation
- **End Users:** Individuals who use the platform to interact and share content

5. User Description

End Users

- Individuals who want to create profiles and share content

- Users who wish to follow others and engage with posts
- Users who expect a simple, responsive, and reliable social media experience

6. Business Requirements

BR-1: User Account Management

- The business requires users to be able to register using valid credentials
- The business requires users to securely log in and log out
- The business requires authentication to access core platform features

BR-2: User Profile Management

- The business requires users to create and maintain personal profiles
- The business requires users to upload profile pictures
- The business requires users to add personal experience or bio information

BR-3: User Interaction

- The business requires users to follow and unfollow other users
- The business requires users to view follower and following counts
- The business requires interaction features such as liking and sharing posts

BR-4: Content Sharing

- The business requires users to create posts containing text and images
- The business requires users to manage (edit/delete) their own posts
- The business requires posts to be displayed in a structured and readable format

BR-5: Personalized Content Feed

- The business requires users to view posts from users they follow
- The business requires the feed to display content in chronological order
- The business requires feed access to be limited to authenticated users

BR-6: Messaging System

- The business requires users to send and receive direct messages

- The business requires users to view message history with other users
- The business requires messages to be accessible only to authorized participants

BR-7: Notification System

- The business requires users to receive notifications for key activities
- Notifications shall be generated for likes, shares, follows, and messages
- The business requires users to view and manage notification status

7. Technology Stack

The following technology stack will be used to develop, deploy, and maintain the Social Media Application:

Backend & Server-Side

- **Backend Framework:** Django (Python)
Used to handle business logic, authentication, and server-side processing.
- **Django ORM:**
Used for database interaction and data management.
- **Django Signals:**
Used to trigger notifications on user actions such as likes, follows, and messages.

Database

- **Relational Database:** PostgreSQL
Used for secure, scalable, and reliable data storage.

Frontend

- **Frontend Styling Framework:** Tailwind CSS
Used to create a modern, responsive, and customizable user interface.
- **Template Engine:** Django Templates
Used for rendering dynamic HTML pages.

Containerization

- **Docker:**
Used to containerize the application to ensure consistent development, testing, and deployment environments.

Development & Version Control

- **Programming Language:** Python
- **Version Control System:** Git
- **Source Code Repository:** GitHub

Deployment & Hosting (Optional)

- **Cloud Deployment:** AWS
- **Alternative Hosting:** PythonAnywhere, Render, DigitalOcean
- **Container Deployment:** Docker-based deployment environment

8. High-Level System Architecture Overview

The Social Media Application will follow a **three-tier web application architecture** to ensure scalability, maintainability, and separation of concerns.

Architecture Layers

1. Presentation Layer

- Handles user interaction and UI rendering
- Built using **HTML**, **Django Templates**, and **Tailwind CSS**
- Responsible for displaying pages such as login, profile, feed, and post views

2. Application Layer

- Implemented using the **Django backend framework**
- Contains business logic, authentication, authorization, and request handling
- Processes user actions such as posting content, following users, and liking posts

3. Data Layer

- Uses **PostgreSQL** as the primary relational database
- Stores user accounts, profiles, posts, relationships, and engagement data
- Accessed through Django ORM to ensure data integrity and security

Containerized Architecture

- The application will be containerized using **Docker**

- Separate containers may be used for:
 - Web application (Django)
 - Database (PostgreSQL)
- Docker ensures consistent environments across development, testing, and deployment stages

High-Level Architecture Flow

1. Users interact with the system through a web browser
2. Requests are sent to the Django application
3. Django processes business logic and authentication
4. Data is retrieved or stored in PostgreSQL
5. The response is rendered and sent back to the user

9. Mapping of Technology Stack to Business Requirements

This section demonstrates how the selected technology stack supports and fulfills the defined business requirements.

BR-1: User Account Management

- **Django:** Handles authentication, session management, and access control
- **PostgreSQL:** Stores user credentials and account information securely
- **Docker:** Ensures consistent authentication behavior across environments

BR-2: User Profile Management

- **Django ORM:** Manages profile data and relationships
- **PostgreSQL:** Stores profile details and experience information
- **Tailwind CSS:** Provides a clean and responsive profile interface

BR-3: User Interaction (Follow, Like, Share)

- **Django:** Implements business logic for follow and engagement actions
- **PostgreSQL:** Stores relationships and interaction data

- **Docker:** Supports scalable deployment as interaction volume grows

BR-4: Content Sharing

- **Django:** Manages post creation, image uploads, and content validation
- **PostgreSQL:** Stores post metadata and content references
- **Django Templates + Tailwind CSS:** Display posts in a user-friendly format

BR-5: Personalized Content Feed

- **Django:** Filters and retrieves posts based on follow relationships
- **PostgreSQL:** Efficiently queries related user content
- **Docker:** Helps maintain performance consistency under varying loads

10. Non-Functional Requirements

The Social Media Application must meet the following non-functional requirements to ensure quality, reliability, and user satisfaction:

10.1 Usability & Responsiveness

- The system shall provide a responsive user interface that functions smoothly across desktop, tablet, and mobile devices.
- The user interface shall be intuitive and easy to navigate for users with basic knowledge of social media platforms.

10.2 Performance

- The application shall load pages within acceptable performance limits under normal usage conditions.
- The system shall handle multiple concurrent users without significant performance degradation.

10.3 Security & Privacy

- User data shall be handled securely and treated as confidential.

- The system shall ensure that only authenticated users can access protected features.
- Users shall only be permitted to access or modify their own data and content.

10.4 Reliability & Availability

- The system shall operate reliably with minimal downtime.
- The application shall handle unexpected errors gracefully without data loss.

11. Success Criteria

The project will be considered successful if the following criteria are met:

- Users can successfully register, log in, and log out of the system.
- Users can create, update, and manage their profiles.
- Users can create, like, and share posts without functional errors.
- Users can view a personalized content feed based on followed users.
- The system demonstrates positive user engagement through regular activity and interaction.
- Users can send and receive direct messages successfully
- Users receive timely notifications for platform activities
- Messaging and notifications increase overall user engagement

12. Assumptions & Constraints

12.1 Assumptions

- Users have stable access to the internet.
- Users are familiar with basic social media concepts such as posting, liking, and following.
- Users access the system using modern web browsers.

12.2 Constraints

- The platform will be developed as a web-based application only.
- The system must comply with applicable data privacy and security regulations.
- Development will be completed within a limited timeframe and available resources.
- The selected technology stack must be used throughout the development lifecycle.

13. Conclusion

This Business Requirements Document (BRD) outlines the business objectives, scope, requirements, and constraints for the Social Media Application. It serves as a foundational reference for system design, development, testing, and validation.

Any modifications to the requirements defined in this document must be formally reviewed and approved by the relevant stakeholders to ensure alignment with business goals and project objectives.

System Requirements Specification

Project: UniVerse (Social Media Platform)

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1. Introduction

1.1 Purpose

The purpose of this SRS is to define the functional and non-functional requirements for the Social Media Platform. This document provides developers, testers, and stakeholders with a clear and comprehensive understanding of the system specifications to ensure a successful implementation.

1.2 Scope

The Social Media Platform allows users to:

- Register and log in securely
- Create and manage personal profiles
- Follow/unfollow users
- Post text and image content
- Like, share, and view posts from followed users
- View a personalized content feed

The system is designed as a **web-based application**, using Django for the backend, PostgreSQL for database management, Tailwind CSS for frontend styling, and Docker for containerization. Optional cloud deployment includes AWS or PythonAnywhere or Render.

1.3 Intended Audience

- Development Team
- Quality Assurance Team
- Project Manager
- Stakeholders/Business Owner

2. Overall Description

2.1 Product Perspective

The Social Media Platform is a **standalone web application**. It is designed using a **three-tier architecture**:

1. **Presentation Layer:** Frontend with Tailwind CSS and Django Templates

2. **Application Layer:** Backend with Django handling business logic and authentication
3. **Data Layer:** PostgreSQL storing user, post, and engagement data

The application will run in **Docker containers** to ensure consistent development and production environments.

2.2 Product Functions

- User registration and authentication
- Profile creation and management
- Follow/unfollow functionality
- Post creation, editing, deletion
- Liking and sharing posts
- Personalized feed displaying posts from followed users
- One-to-one direct messaging between users
- In-app notification delivery and management

2.3 User Classes and Characteristics

User Type	Description
End User	Creates content, follows users, likes/shares posts
Admin/Manager	Monitors the system, reviews usage metrics (future implementation)

2.4 Operating Environment

- Web browsers: Chrome, Firefox, Edge, Safari
- Devices: Desktop, tablet, mobile
- Server OS: Linux (Ubuntu recommended)
- Docker containerized environment

2.5 Design & Implementation Constraints

- Must use Django, PostgreSQL, Tailwind CSS
- Must comply with data privacy regulations

- Web-based only; mobile app not included in this version

2.6 Assumptions and Dependencies

- Users have internet access
- Users are familiar with social media basics
- Future scalability may require load balancing or cloud optimization

3. Functional Requirements

3.1 User Account Management

- FR-1.1: Users shall register with valid credentials (email/username and password)
- FR-1.2: Users shall log in and log out securely
- FR-1.3: Users shall access core features only when authenticated

3.2 Profile Management

- FR-2.1: Users shall create, update, and manage profiles
- FR-2.2: Users shall upload profile pictures
- FR-2.3: Users shall add personal information, experience, or bio

3.3 Social Interaction

- FR-3.1: Users shall follow/unfollow other users
- FR-3.2: Users shall view follower and following counts
- FR-3.3: Users shall like and share posts

3.4 Content Sharing

- FR-4.1: Users shall create posts containing text and images
- FR-4.2: Users shall edit or delete their own posts
- FR-4.3: Posts shall be displayed in structured, readable format

3.5 Personalized Feed

- FR-5.1: Users shall view posts from followed users in chronological order
- FR-5.2: Only authenticated users shall access the feed

3.6 Messaging System

- FR-6.1: Users shall send direct messages to other users
- FR-6.2: Users shall receive messages in real time or near real time

- FR-6.3: Users shall view chat history with specific users
- FR-6.4: Only authenticated users shall access messaging features

3.7 Notification System

- FR-7.1: System shall generate notifications for likes, shares, follows, and messages
- FR-7.2: Users shall view notifications in a dedicated notification section
- FR-7.3: Notifications shall indicate read and unread status
- FR-7.4: Only authenticated users shall receive notifications

4. Non-Functional Requirements

4.1 Usability

- System shall have a responsive interface on desktop, tablet, and mobile devices
- User interface shall be intuitive and easy to navigate

4.2 Performance

- Page loading times shall remain within 2–3 seconds under normal load
- System shall support concurrent users initially
- Notifications shall be delivered within acceptable time limits
- Messaging system shall handle concurrent conversations without delays

4.3 Security

- Passwords shall be stored securely using hashing
- Only authenticated users can perform protected actions
- Users can only access or modify their own data

4.4 Reliability & Availability

- System shall handle errors gracefully without data loss
- Downtime shall be minimized; system shall be highly available

4.5 Scalability

- Docker containers shall allow easy scaling
- PostgreSQL design shall support increasing user and post volume

5. System Architecture

- **Frontend:** HTML + Tailwind CSS + Django Templates
- **Backend:** Django framework
- **Database:** PostgreSQL
- **Containerization:** Docker (web app and database in separate containers)
- **Optional Deployment:** AWS / PythonAnywhere / Render / DigitalOcean

High-Level Flow:

1. User interacts with web browser
2. Request sent to Django backend
3. Business logic processed, database queried via Django ORM
4. Response rendered via Django Templates
5. Result displayed in browser

6. Database Requirements

6.1 User Table

- user_id (PK)
- username
- email
- password_hash
- profile_picture_url
- bio

6.2 Post Table

- post_id (PK)
- user_id (FK)
- text_content
- image_url
- timestamp

6.3 Follow Table

- follower_id (FK)
- following_id (FK)

6.4 Likes Table

- user_id (FK)
- post_id (FK)

6.5 Shares Table

- user_id (FK)
- post_id (FK)

6.6 Message Table

- message_id (PK)
- sender_id (FK)
- receiver_id (FK)
- message_content
- timestamp
- is_read

6.7 Notification Table

- notification_id (PK)
- user_id (FK)
- notification_type
- reference_id
- is_read
- timestamp

7. Technology Stack Mapping

Requirement	Technology
Authentication	Django, PostgreSQL, Docker
Profile Management	Django ORM, PostgreSQL, Tailwind CSS
Content Posting	Django, PostgreSQL, Django Templates
Feed & Social Interaction	Django, PostgreSQL, Docker
UI/UX	Tailwind CSS, Django Templates
Deployment	Docker, AWS/PythonAnywhere

8. Success Criteria

- Users can register, log in, and log out without issues
- Users can create, edit, delete posts and profiles
- Users can follow, like, and share posts
- Personalized feed displays posts correctly
- System performance, security, and usability requirements met

9. Assumptions & Constraints

Assumptions:

- Users have access to modern web browsers and internet
- Users understand social media basics

Constraints:

- Web-based only
- Data privacy regulations must be followed
- Technology stack (Django, PostgreSQL, Tailwind, Docker) must be used
- Project must be completed within the planned timeline

10. Future Enhancements

- Search and filtering of posts
- Analytics dashboard for user engagement

11. Conclusion

This System Requirements Specification (SRS) defines the functional and non-functional requirements, system architecture, database design, and technology stack for the Social Media Platform.

It serves as a comprehensive reference for the development and quality assurance teams to ensure that the system meets the business objectives defined in the BRD.

Any modifications to the requirements specified in this document must be formally reviewed and approved by the relevant stakeholders to ensure alignment with business goals and project constraints.